Rotational Dynamics

Building Models to Describe our World

Describing angular statics and dynamics

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- Torque

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Angular Quantities and Torque

Describing Angular Statics and Dynamics

Context

We started with $\sum \vec{F} = m\vec{a}$, and then:

- Used $\sum \vec{F} = m\vec{a}$ as is.
- Took $\int (\sum \vec{F} = m\vec{a}) \cdot d\vec{l}$ to define work and energy.
- Took $\int (\sum \vec{F} = m\vec{a}) dt$ to introduce impulse and momentum.

This week, we do:

$$\vec{r} \times \left(\sum \vec{F} = m\vec{a} \right)$$

in order to model rotating systems.

Describing things that rotate in 2D

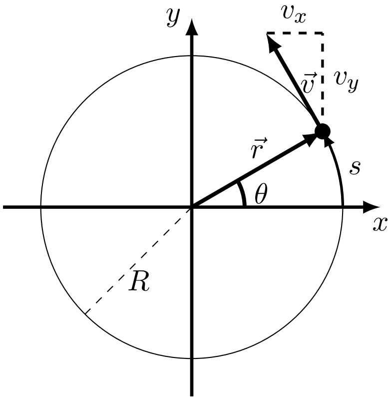
- Recall that we can describe circular motion using linear or angular quantities:

$$s = R\theta$$
$$v = R\omega$$
$$a_T = \alpha R$$

- The angular quantities are **related to the tangential** linear quantities. The acceleration also has a radial component:

$$a_R = \frac{v^2}{R} = \omega^2 R$$

due to the **change of the direction** of the velocity vector (not its magnitude).

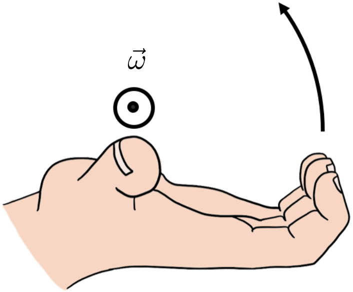


Variables to describe circular motion.

Describing things that rotate in 3D

- In general, things rotate in a circle:
  - About an axis (defining a normal plane).
  - In a specific direction around the axis.
  - With a specific amplitude/speed/acceleration.
- Use "axial/pseudo vectors" to specify:
  - $\vec{\theta}$, angular displacement.
  - $\vec{\omega}$, angular velocity.
  - $\vec{\alpha}$, angular acceleration.

  with magnitudes given by the same value as in 2D, and **direction given by the "right hand rule for axial vectors"**.



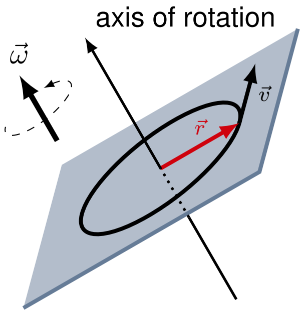
The right-hand rule for axial vectors is used to define the direction of a vector associated with rotation. The thumb is parallel to the axis of rotation and the fingers curl in the direction of rotation.

Angular vectors from linear vectors

- Need to be able to describe rotation about any axis (not just x, y, z).
- The angular vectors are related to the linear ones:

$$\vec{\omega} = \frac{1}{r^2}\vec{r} \times \vec{v}$$
$$\vec{\alpha} = \frac{1}{r^2}\vec{r} \times \vec{a}$$

- The vector, $\vec{r}$, is perpendicular to the axis of rotation and goes **from the axis of rotation to the particle's position**.



An object rotating in a circle in an arbitrary plane. We can define all angular quantities using the linear ones and the vector $\vec{r}$ from the axis of rotation to the position.

Torque

- Recall: A net force exerted on an object causes it to accelerate – forces were introduced as a mathematical quantity to help model the causes of motion.

$$\sum \vec{F} = m\vec{a}$$

- Torque is analogous to force, but torques cause an object to have an angular acceleration **about a specific axis**:
  - A torque is **only defined relative to a particular axis.**
  - If there is a net torque on an object, it will have an angular acceleration about that axis:

$$\sum \vec{\tau} = mr^2\vec{\alpha}$$

$\rightarrow$ A larger torque will result in a larger angular acceleration.

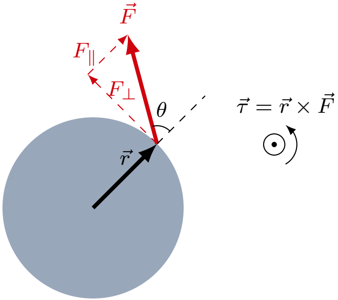
Torque from a force

- Measures how much a force will cause an object to rotate about a given axis (this determines $\vec{r}$).
- Defined as:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

with magnitude:

$$\tau = rF\sin\theta$$



The torque from a force.

Torque from a force

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  with magnitude:

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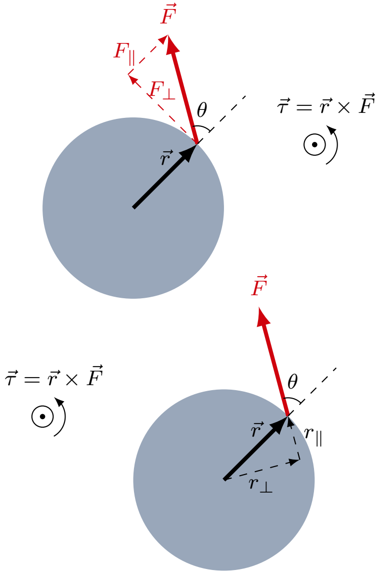
- The cross product "picks out" the components of the vectors that are perpendicular to each other:

$$\tau = rF_{\perp} = r_{\perp}F$$

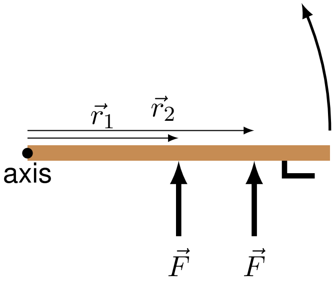


The torque from a force.

Torque to rotate a door

$$\vec{\tau} = \vec{r} \times \vec{F}$$
$$\tau = rF \sin\theta$$

- A given force will result in the largest torque if it is perpendicular to the vector, $\vec{r}$ $(\sin\theta = 1)$.
- A given force will result in the largest torque if the vector, $\vec{r}$, has the largest magnitude.
- That's why the door handles are far from the hinges... You knew this intuitively...



Rotating a door is easier if you push far away from the hinges, right?

Angular Dynamics and Moment of Inertia

Describing Angular Statics and Dynamics

Dynamics of rotating particles

- For a point particle:

$$\vec{F}^{net} = m\vec{a}$$

- If we choose an axis of rotation, such that, $\vec{r}$, is the vector from that axis to the particle:

$$\vec{r} \times \vec{F}^{net} = m\vec{r} \times \vec{a} \qquad \text{recall: } \vec{\alpha} = \frac{1}{r^2}\vec{r} \times \vec{a}$$

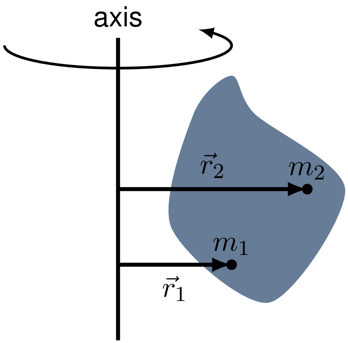
$$\therefore \vec{\tau}^{net} = mr^2\vec{\alpha}$$

- This **equation only makes sense relative to a specific axis of rotation**. It is valid even if there is nothing actually rotating!
- The inertia of the particle to rotational motion is given by mr^2, and **depends on both the mass of the particle and its distance to the axis of rotation**.
- We often refer to this as the rotational version of Newton's Second Law.

Dynamics of rotating objects

- We can extend the description of how one particle rotates about an axis to how a solid object rotates about an axis.
- An object can be modeled as a "system" of particles, each with a torque exerted on it, and a corresponding angular acceleration:

$$\vec{\tau}_1^{net} = m_1 r_1^2 \vec{\alpha}_1$$
$$\vec{\tau}_2^{net} = m_2 r_2^2 \vec{\alpha}_2$$
$$\dots$$



Two particles in a solid object rotating about an axis.

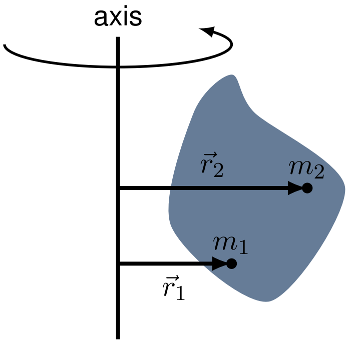
Dynamics of rotating objects

- Rotational version of Newton's Second Law true for each particle:

$$\vec{\tau}_1^{\,net} = m_1 r_1^2 \vec{\alpha}_1$$
$$\vec{\tau}_2^{\,net} = m_2 r_2^2 \vec{\alpha}_2$$
$$\dots$$

- Sum these all together:
  - **All particles have the same angular acceleration.**
  - When summing the torques on each particle, all of the internal torques (due to internal forces) cancel:

$$\sum \vec{\tau}^{\,ext} = \left(\sum m_i r_i^2\right) \vec{\alpha}$$



The particles in a solid object rotating about an axis all have the same angular acceleration relative to that axis.

The moment of inertia

- The net external torque on an object determines its angular acceleration:

$$\sum \vec{\tau}^{ext} = \left(\sum m_i r_i^2\right) \vec{\alpha}$$

- We introduce, I, the "moment of inertia" of an object **relative to a given axis of rotation**:

$$I = \sum m_i r_i^2$$

- Rotational version of Newton's Second Law:

$$\sum \vec{\tau}^{ext} = I\vec{\alpha}$$

- Instead of force, we have torque, instead of mass, we have moment of inertia, instead of linear acceleration, we have angular acceleration.

Rotational vs Linear dynamics

- Recall, for an object, the **net external force** on the object determines **the acceleration of the CM**:

$$\sum \vec{F}^{ext} = M\vec{a}_{CM}$$

- Similarly, the **net external torque** about an axis on the object determines its **angular acceleration** relative to that axis:

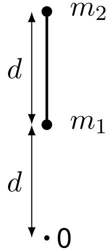
cause of motion $\qquad \sum \vec{\tau}^{ext} = I\vec{\alpha} \qquad$ (resistance to motion)(kinematic quantity)

- The moment of inertia has a similar "role" as the mass in Newton's Second Law. For a given torque, the object with the smaller moment of inertia will have the largest angular acceleration.

Group exercise

- Calculate the moment of inertia for a system of 2 masses connected by a mass-less rod of length d, for:
  - An axis that goes through m_1.
  - An axis that goes through 0 (d away from m_1). In both cases, the axis is perpendicular to the page.

$$I = \sum m_i r_i^2$$



Two masses connected by a mass-less rods rotate around an axis perpendicular to the page.

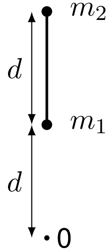
Group exercise

- Calculate the moment of inertia for a system of 2 masses connected by a mass-less rod of length d, for:
  - An axis that goes through m_1.
  - An axis that goes through 0 (d away from m_1). In both cases, the axis is perpendicular to the page.

$$I = \sum m_i r_i^2$$

- Applying the formula:

$$I_{m_1} = m_1(0)^2 + m_2 d^2 = m_2 d^2$$
$$I_0 = m_1(d)^2 + m_2(2d)^2 = (m_1 + 4m_2)d^2$$



Two masses connected by a mass-less rods rotate around an axis perpendicular to the page.

Calculating moment of inertia

- Similar to centre of mass, which is a mass-weighted sum of the position coordinate:

$$x_{CM} = \frac{1}{M} \sum m_i x_i = \frac{1}{M} \int_0^M x dm$$

- For the moment of inertia, it is a mass-weighted sum of the **distance to the axis of rotation squared**:

$$I = \sum m_i r_i^2 = \int_0^M r^2 dm$$

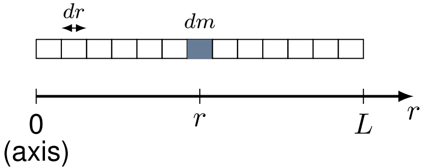
Continuous objects: Moment of inertia of a rod about its end

- Define mass per unit length: $\lambda = \frac{M}{L}$
- Choose a mass element on the rod, dm, of length, dr , at a distance, r, from the axis of rotation, it has mass:

$$dm = \lambda dr$$

- The total moment of inertia is the sum over all mass elements:

$$I_{end} = \sum m_i r_i^2 = \int_0^M r^2 dm = \int_0^L \lambda r^2 dr$$
$$= \lambda \frac{1}{3} L^3 = \frac{1}{3} M L^2$$



To find the moment of inertia of a one dimensional object, model it as being made of many point masses, dm, each with an infinitesimal length, dx.

Group exercise

Determine the moment of inertia of a uniform rod about is centre of mass:

- Do you expect it to be bigger or smaller than moment of inertia about its end?

Continuous objects: Moment of inertia of a rod about its end

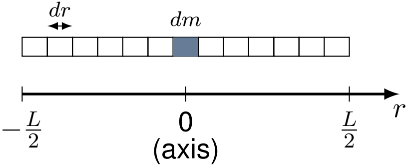
- Define mass per unit length: $\lambda = \frac{M}{L}$
- Choose a mass element on the rod, dm, of length, dr , at a distance, r, from the axis of rotation, it has mass:

$$dm = \lambda dr$$

- The total moment of inertia is the sum over all mass elements:

$$I_{CM} = \sum m_i r_i^2 = \int_0^M r^2 dm = \int_{-\frac{L}{2}}^{+\frac{L}{2}} \lambda r^2 dr$$

$$= \lambda \left[\frac{1}{3} r^3\right]_{-\frac{L}{2}}^{+\frac{L}{2}} = \frac{1}{12} \lambda M L^3 = \frac{1}{12} M L^2$$



To find the moment of inertia of a rod about CM, we move the origin to the axis of rotation, so that r measures distance to the axis.

Alternatively, we could have added the moment of inertia of two rods of length $\frac{1}{2}L$ (and mass $\frac{M}{2}$) rotated about their ends:

$$I_{CM} = 2\frac{1}{3}\frac{M}{2}\left(\frac{L}{2}\right)^2 = \frac{1}{12} M L^2$$

Parallel axis theorem

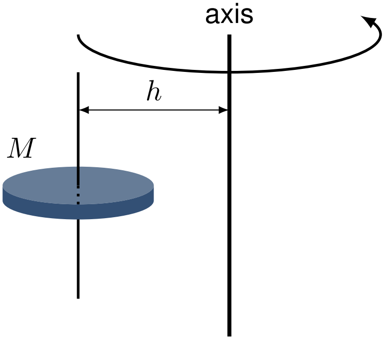
- Given the moment of inertia about an axis that goes through the CM, the moment of inertia about an axis that is parallel and goes through a different point, a distance, h away is given by:

$$I_h = I_{CM} + Mh^2$$

- For a rod, find the moment of inertia about the CM, given the moment of inertia about one end:

$$I_{end} = \frac{1}{3}ML^2$$

$$I_{CM} = I_{end} - M\left(\frac{L}{2}\right)^2 = \frac{1}{3}ML^2 - \frac{1}{4}ML^2 = \frac{1}{12}ML^2$$



The moment of inertia of an object about an axis a distance, h, away from the CM is given by the parallel axis theorem.

Quick review

- Torque from a force **about an axis**:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

- Rotational equivalent of Newton's Second Law:

$$\vec{\tau}^{net} = mr^2\vec{\alpha} \quad \text{(particle)}$$
$$\sum \vec{\tau}^{ext} = I\vec{\alpha} \quad \text{(object)}$$

- Moment of inertia of an object **about an axis**:

$$I = \sum m_i r_i^2 = \int_0^M r^2 dm$$

- Parallel axis theorem:

$$I_h = I_{CM} + Mh^2$$

quilibriem with Rotational Dynamics

Describing Angular Statics and Dynamics

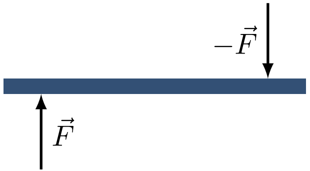
Equilibrium

- Just because the sum of the external forces on an object is zero does not mean that the object does not move. It only means that the centre of mass does not accelerate:

$$\sum \vec{F}^{ext} = m\vec{a}_{CM}$$

- For the object to be "in equilibrium" the sum of the torques on the object must also be zero:

$$\sum \vec{\tau}^{ext} = I\vec{\alpha}$$



The net force on this bar is zero, but it will rotate because the net torque (about any axis!) is not zero.

Equilibrium

- Just because the sum of the external forces on an object is zero does not mean that the object does not move. It only means that the centre of mass does not accelerate:

$$\sum \vec{F}^{ext} = m\vec{a}_{CM}$$

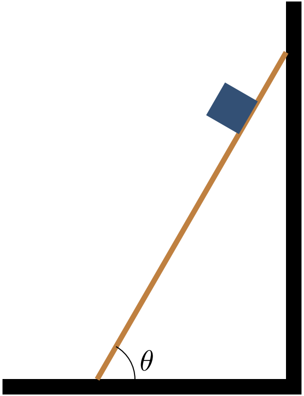
- For the object to be “in equilibrium” the sum of the torques about any axis on the object must also be zero:

$$\sum \vec{\tau}^{ext} = I\vec{\alpha}$$

The net force on this bar is zero, but it will rotate because the net torque is not zero. Note that it will rotate about the CM because that is the point that behaves according to the net external force, which is zero, so the CM does not move!

Example: ladder

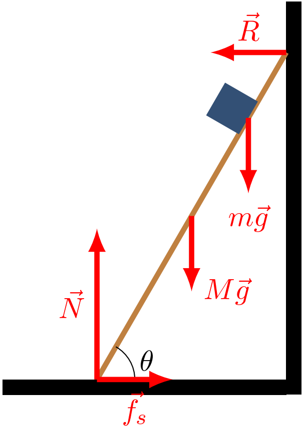
- Group question: Draw the forces on the ladder.



A ladder with a weight on it.

Example: ladder

- The friction force on the ground prevents the bottom of the ladder from sliding away.
- If you consider the torques about the contact point with the ground, the torque on the ladder from the weight of the person on the ladder gets bigger as they climb up:
  - Only R can grow and exert a torque in the opposite direction to keep the ladder in equilibrium.
  - As R increases, f_s must also increase (not net force in the horizontal direction).
- When someone "spots you" on a ladder, the most important is that they prevent the bottom of the ladder from sliding away from the wall.

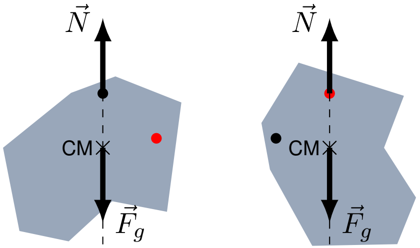


Forces exerted ***on the ladder****.*

Ladder safety: which is safest (from a statics perspective)?

Example: Find CM of object

- Only two forces on the object:
  - Weight **exerted at the CM.**
  - Normal force exerted by the pin.
- Since the object is in equilibrium, there is no net torque.
- The CM must be located exactly below the pin, or the weight and normal force would create a net torque.



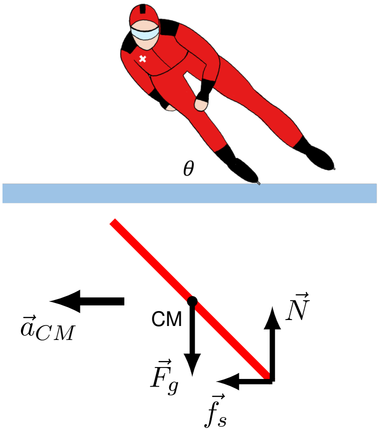
Rotating a 2D shape and using a plumbline to find its CM. The CM corresponds to the location where the net force of gravity is exerted, sometimes called the "centre of gravity".

Dynamic vs Static equilibrium

- Equilibrium $\rightarrow$ Sum of the torques is zero.
- Static equilibrium $\rightarrow$ The net force is zero (the CM does not accelerate).
- Dynamic equilibrium $\rightarrow$ The net force is not zero (the CM accelerates, as does any axis of rotation that is “attached” to the object).

Speed skater in dynamic equilibrium

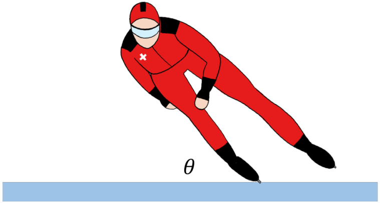
- The skater is accelerating to the left (going around a curve, centre of the circle is to the left).
- The skater is **not** rotating about her CM!
    - The torques from the normal force and the friction force cancel.
- However, taking the torques about the bottom point, we find there is a net torque from gravity. What is going on?

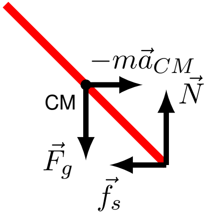


Forces on a speed skater, as viewed from the ground.

Speed skater in dynamic equilibrium

- The sum of torques about the bottom point is not zero! Why?
  - The accelerating skater is a **non-inertial frame of reference.** The **bottom point corresponds to an axis that is accelerating** with the skater.
  - You can take the torques about the bottom point if you consider **inertial forces.**
  - You **can always take the torques about the CM**, because the inertial force exerted at the CM will result in no torque, so it won't matter if the CM is accelerating.

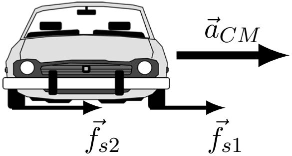




Forces on a speed skater, as viewed from a reference frame accelerating with the skater. You need to include inertial forces if taking torques in an accelerating frame of reference!

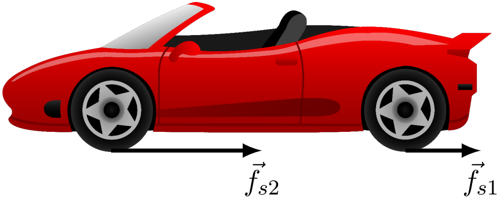
Car going around a curve

- The friction forces, $\vec{f}_s$, from the tires, perpendicular to the direction of motion, provide the "centripetal force" (the net force towards the centre of the circle).
- These create a net torque about the CM of the car.
- The car leans to the left (and compresses the shocks which provide the counter torque).



Forces on a cornering car.

Braking



Due to the torque from the braking wheels and resulting normal force, the front wheel(s) do most of the braking

- The friction forces from the tires provide the force to decelerate.
- The front of the car goes down because of the net torque about the CM of the car.
- The normal force is greater at the front, the front wheels do most of the breaking! Look at the breaks in a car/motorcycle: they are *way* bigger on the front wheels!